(which can fit in 1 or 2 boxes)

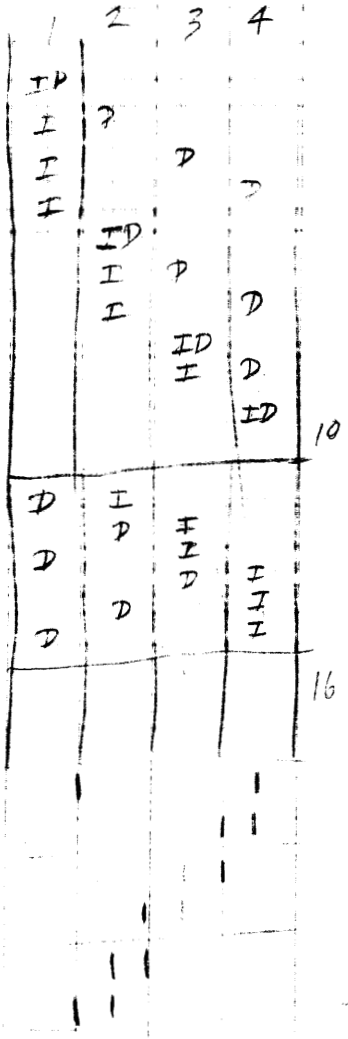
1. Assume one simple program in machine, all in memory at once,
2. Data and instructions can be separate but are fetched
   [illegible] call it one [illegible] I + D fetch or I fetch, D fetch separately,
3. [illegible]
4. [illegible]
   4 [illegible], or (b) 2, 2
   6 [illegible], (b) 2 + 4, or [illegible]
   8 [illegible], (b) 2+6, (c) 4+4, (d) 2, 2, 4, or (e) 2, 2, 2, 2
   10 [illegible], (b) 2+8 (c) 4+6, (d) 2, 2, 6, (e) 2, 4, 4, (f) 2, 2, 2, 4
   [illegible] 2, 2, 2, 2, 2
   12 [illegible], (b) [illegible]

Table of Speeds on Mod Prob. for various Memory combinations

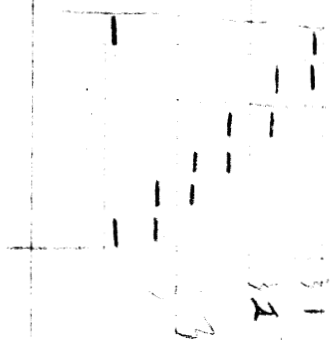
	Jan '58 STD Time (100× speed)	Apr '58 Time (72× speed)	Aug '58 Time (60× speed)
(1) 1 memory box.			
(a) [illegible]	~52	mixed Random 42	39
(2) 2 Memory boxes			
(a) [illegible]	68	mixed Random 57	51
(3) 4 Memory boxes			
(a) [illegible]	82	mixed [illegible]	56
(b) 2, 2	Random = 81, Sp = 13	Random = 64, Sp = 70	Random = 56, Sp = 61
(c) 2+2	68	57	51
(d) 2+, 2	73	70	61

2, 2, 2 cases

2_{ID}	$4+4+4=12$
$2_I, 2_D$	24
	36

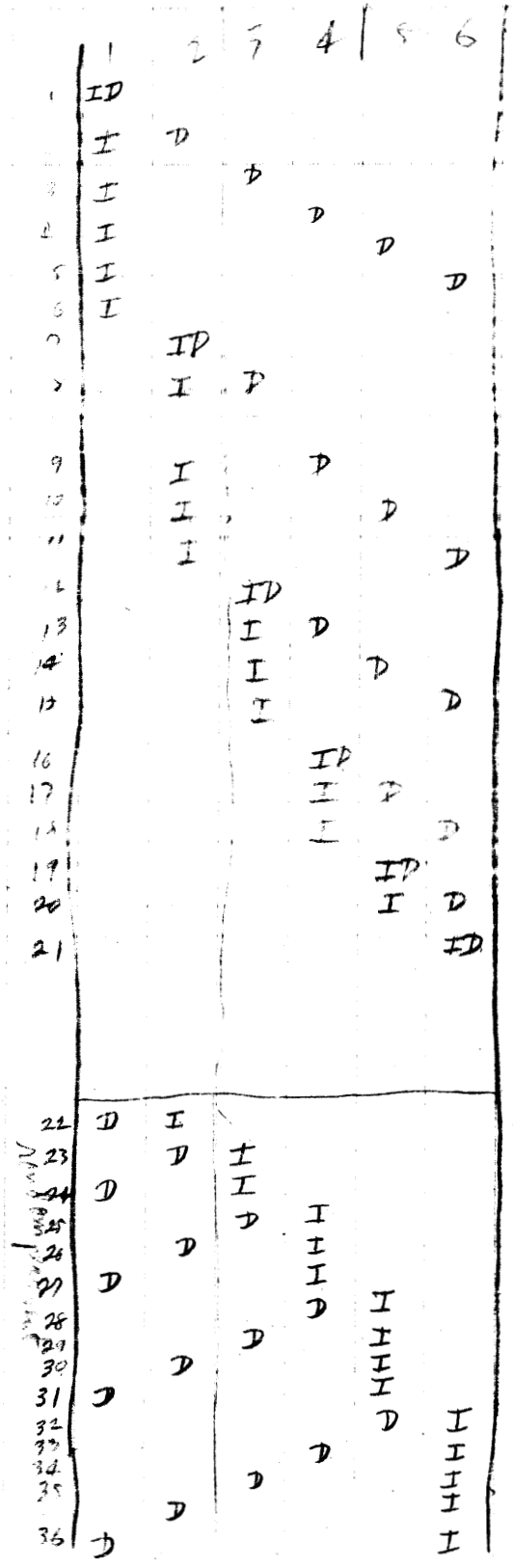


	all cases	omitting DI
$2I+D$ =	8	8
$2I+2D$ =	16	4



2-4 cases

	allowing all cases	omitting DI
$2I+D$ =	4	3
$2I+4D$ =	8	8
$4I+D$ =	16	10
$2D+I$ =	8	0
	36	21



These omitted for DI cases forbidden

Example

$(14)(68)$ =	272
$(8)(100)$ =	800
$(16)(82)$ =	1312
$(8)(93)$ =	744
	3128
	86.9

$(3)(68)$ =	204
$(8)(100)$ =	800
$(10)(82)$ =	820
	1824
	86.9

$(4)(51)$ =	
$(8)(61)$ =	
$(16)(56)$ =	
$(8)(61)$ =	
	57.7

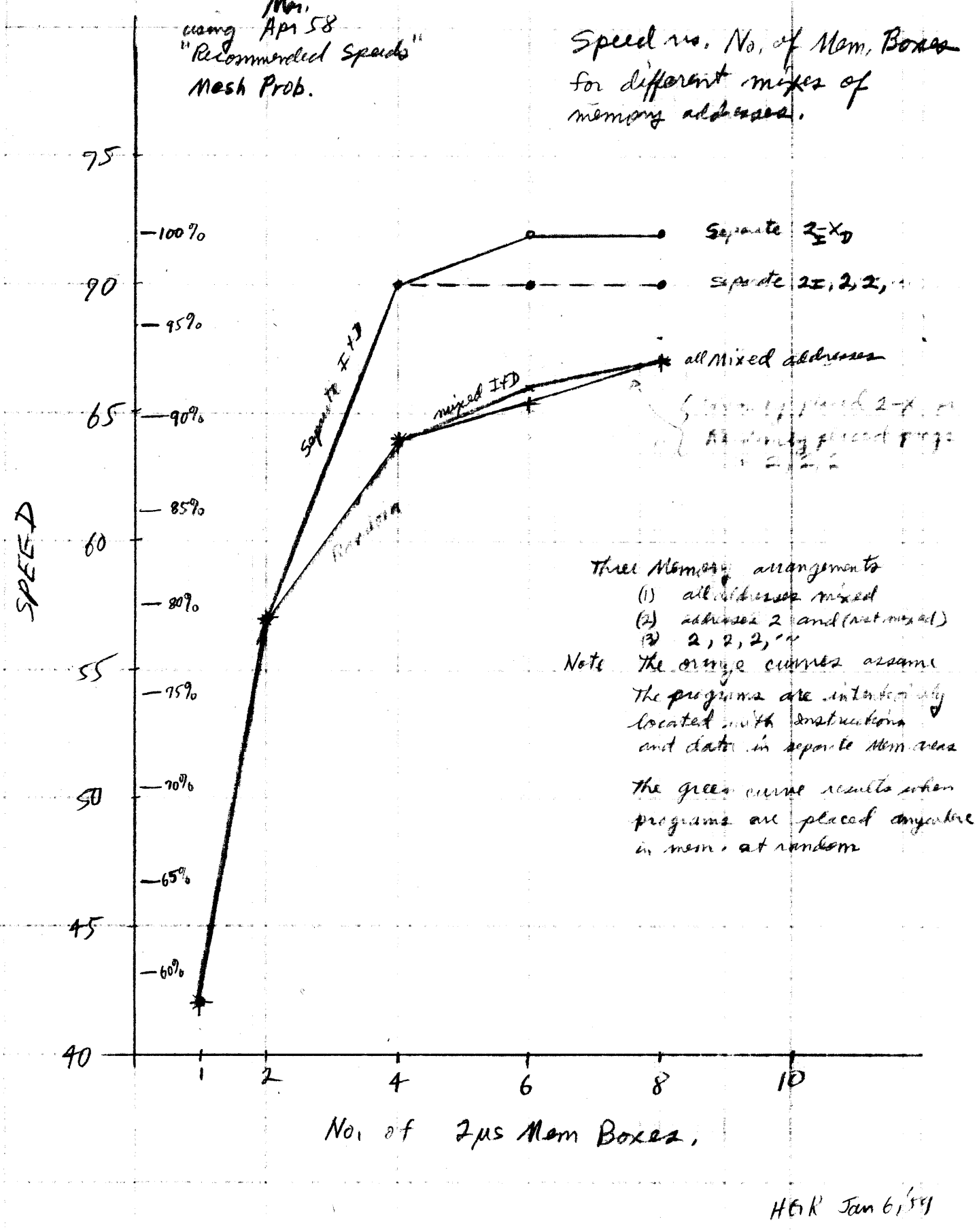
$(3)(51)$ =	
$(8)(61)$ =	
$(10)(56)$ =	
	57.2

		Jan '57 STD Times (10 × Speed)	Apr '58 Rec Times (72 × Speed)	Sept '58 Times (55 × Speed)
	(3+1)			
	(4) 6 [illegible]			
	(a) [illegible]	85	mixed = 65.	59.
(weight)	(b) 2 and 4	Random = ~~88~~ 87., Sep = 100	Random = ~~66~~ 66., Sep = 72.	Random = ~~58~~ 58., Sep = 61
.111	✓ (i) $2_I \times 1_D$ [illegible]	68	57.	51.
.222	✓ (ii) $2_I \times 4_D$ sep	100	72.	61.
.222	[illegible] (i) $2_D \times 4_I$ ($= 1^{-1}$)	93	70.	61
.445	✓ (ii) 4 + 4D [illegible]	82	64.	56
	(c) 2, 2, 2	Random = ~~85~~, Sep = 95	Random = ~~66~~, Sep = 70.	Random = ~~58~~, Sep = 61
.333	✓ (i) 2_{I+D} [illegible]	68	57.	51.
.667	[illegible] 2_D sep	93	70.	61
	(5) 8 [illegible]			
	(a) [illegible]	88	67.	57.
	(b) 2 and 6	Random = ~~89~~ 89., Sep = 103	Random = ~~67~~ 67., Sep = 72	Random = ~~59~~ 59., sep = 61
.0625	✓ (i) 2_{I+D}	68	57.	51.
.5625	✓ (ii) 6_{I+D}	85	65.	59
.1875	✓ (iii) $2_I \times 6_D$ sep	103	72.	61
.1875	(iv) $2_D \times 6_I$ sep	93	70.	61
	(c) 4 and 4	Random = 61, Sep = 110	Random = 68, Sep = 72	Random = 57, Sep = 61
	✓ (i) 2_{I+D}	82	64.	56
	✓ (ii) $4_I \times 4_D$ sep	100	72.	61
	(d) 2, 2, 4	Random = 86, sep = 100	Random = 66, Sep = ~~72~~ 70	Random = 57, sep = 61
	✓ (i) [illegible]	68	57.	51
	✓ [illegible] $2_I \times 2_D$ [illegible]	93	70.	61
	✓ (iii) 4_{I+D}	82	64.	56
	(iv) $4_I \times 2_D$ [illegible]	93	70.	61
	✓ [illegible] $2_I \times 4_D$ [illegible]	100	72.	61

weight (e)	2, 2, 2, 2	Random = ~~87~~ 87. Sep=93	Random = ~~64~~ 67. Sep=70	Random = ~~X~~ 59. Sep=61
(.25)	$r(e) = 2_I$	68	57.	51.
(.75)	$r(e) = 2_I + 2_D$	93	70.	61

Num comp 2 = 4, ~~6~~, ?

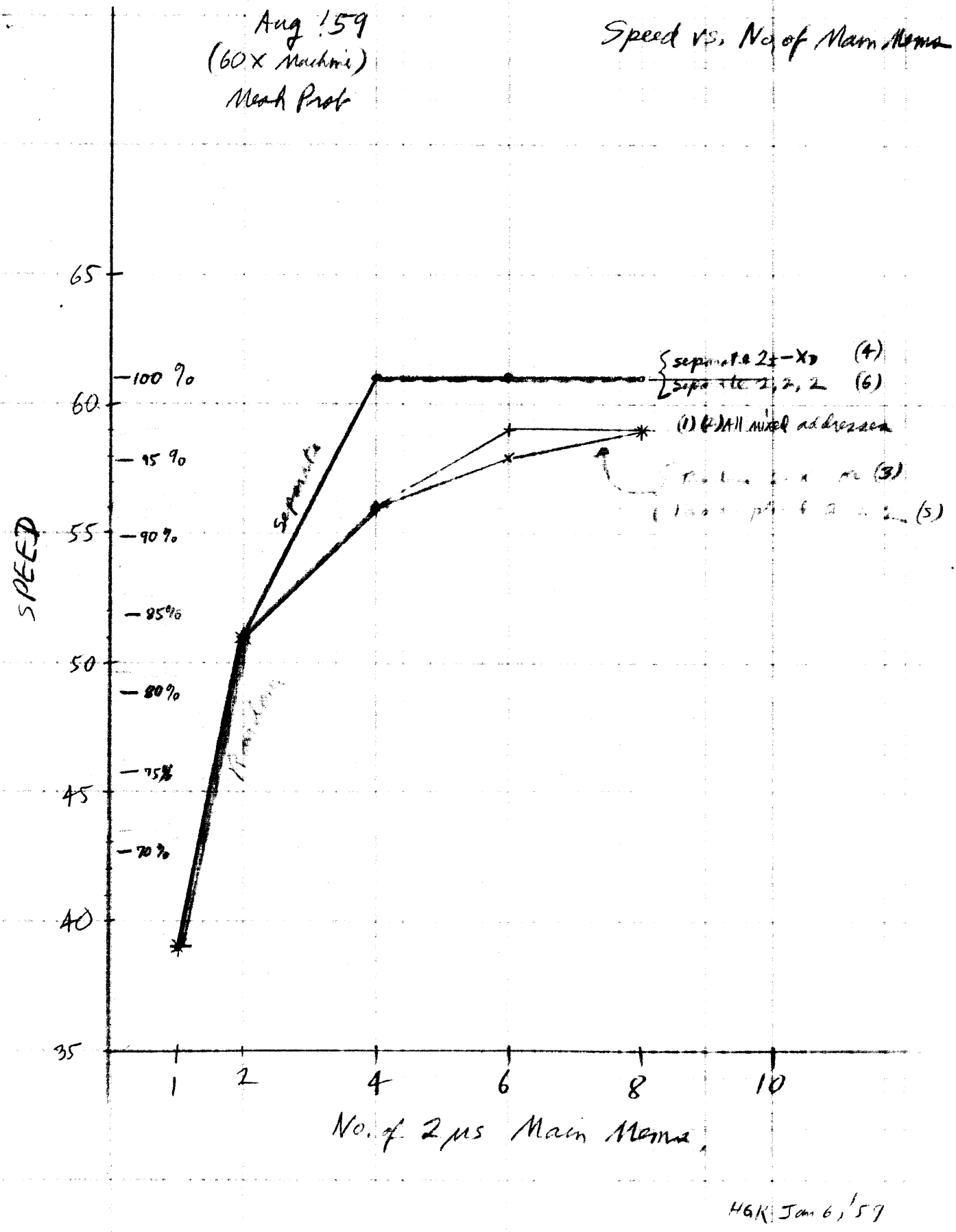
two decoders		total
2		2
2	2	4
2	4	6
4	4	8
2	8	10
4	8	12
8	8	16


Speed vs. No. of Mem. Boxes for different mixes of memory addresses.
using Apr 58 "Recommended Speeds" Mesh Prob.
75
100%
90
95%
65
90%
85%
60
80%
55
75%
50
70%
65%
45
60%
40
SPEED
Separate I+D
mixed I+D
Random
Separate Σ-X_D
Separate 2Σ, 2, 2, ...
all Mixed addresses
Three Memory arrangements
(1) all addresses mixed
(2) addresses 2 and (not mixed)
(3) 2, 2, 2, ...
Note The orange curves assume The programs are intentionally located with instructions and data in separate Mem areas
The green curve results when programs are placed anywhere in mem. at random
1
2
4
6
8
10
No. of 2μs Mem Boxes.

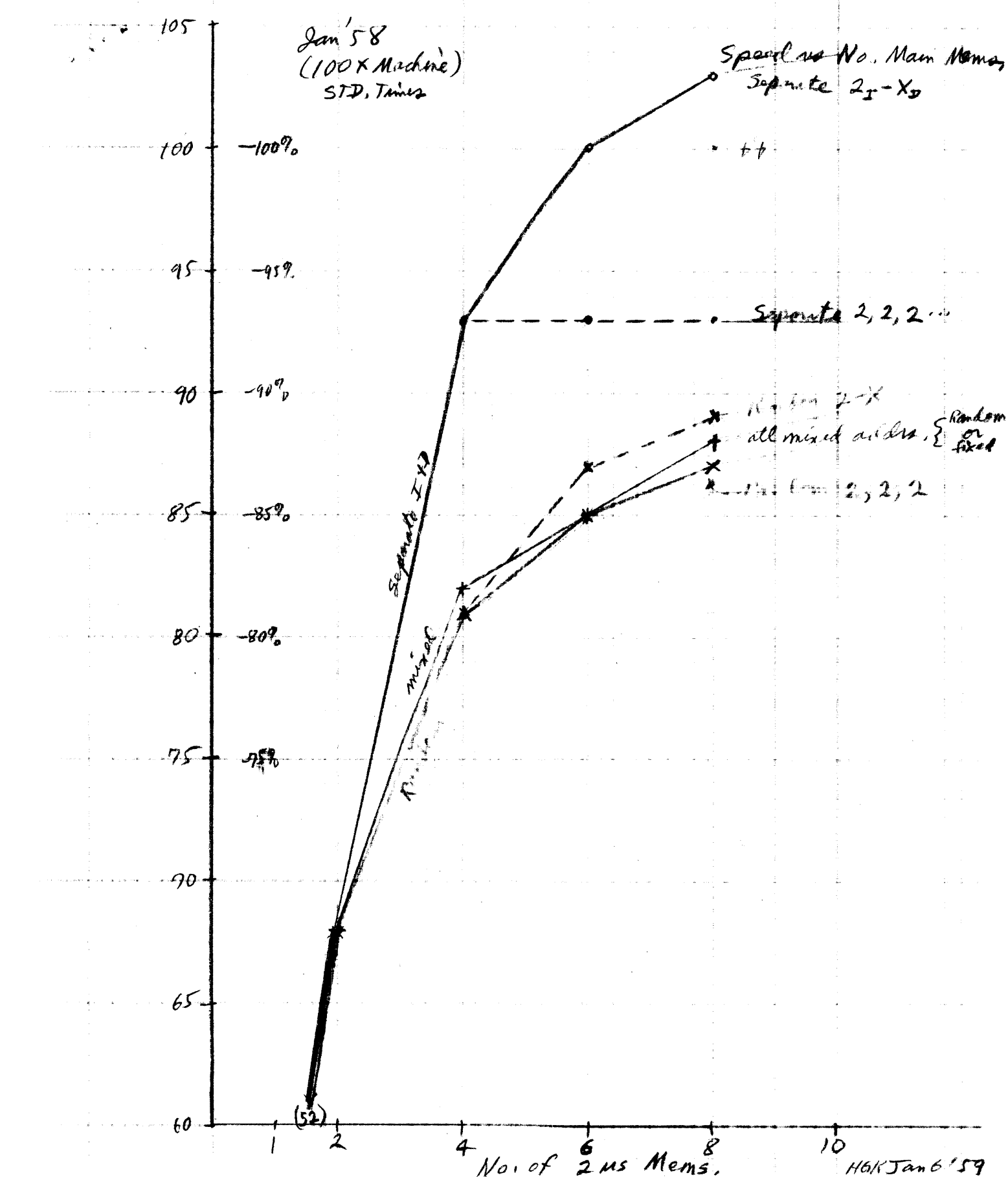
HGR Jan 6, '59

Aug '59
(60 × Machine)
Mesh Prob

Speed vs. No of Main Memo



HGK Jan 6, '59


Jan '58
(100 × Machine)
STD. Times
Speed vs No. Main Mems
Separate $2_I - X_D$
Separate 2, 2, 2
all mixed addrs, { Random or fixed
Separate I+D
mixed
105
100
95
90
85
80
75
70
65
60
100%
95%
90%
85%
80%
75%
(52)
1
2
4
6
8
10
No. of 2 μs Mems.
HGK Jan 6 '59